Multiple IV Bolus Practice 3

Module 6 - Multiple Dosings · Daily practice sheet · No date

Name _____ Date _____

A 80 kg patient is scheduled to receive a 20 mg/kg IV bolus injection of a medication every 6 hours. The drug has an apparent volume of distribution that is 25% of body weight and a total body clearance is 4.62 L/hr. Determine the following for this medication in this patient:

a Elimination half-life.

b Maximum plasma drug concentration of the first dose.

c Minimum plasma drug concentration of the first dose.

d Maximum steady-state plasma drug concentration.

e Minimum steady-state plasma drug concentration.

f Average steady-state plasma drug concentration.

g Average amount of drug in the body during steady state.

h Plasma level 12 hours after the last dose, assuming steady-state levels are achieved.

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

a

Her key

Answer: 3 hr

WHAT THE STEM GIVES

weight	80 kg	$V_D = 0.25 \times 80 = 20$ L
D_0	20 mg/kg	*not needed: the half-life depends on V_D and Cl_T alone*
τ	every 6 hours	*not needed*
V_D	25% of body weight	0.25 L per kg × 80 kg = 20 L, the numerator
Cl_T	4.62 L/hr	the denominator

THE WORKING, ONE LINE AT A TIME

UNIT

$$V_D = (0.25\ \text{L/kg})(80\ \text{kg}) = 20\ \text{L}$$

A volume given as 25% of body weight is read as 0.25 L per kilogram, so for an 80 kg patient it is 0.25 × 80 = 20 L; the clearance is already in L/hr, so the two units match.

SETUP

$$t½ = \frac{0.693\ V_D}{Cl_T}$$

$Cl_T = k\ V_D$ and $k = 0.693$ over $t½$ combine to $t½ = 0.693\ V_D$ over Cl_T, so the half-life comes straight from the volume and the clearance without finding k first.

ALGEBRA

$$t½ = \frac{(0.693)(20\ \text{L})}{4.62\ \text{L/hr}} = 3\ \text{hr}$$

Litres cancel between the volume and L/hr, leaving hours; $k = \frac{4.62\ \text{L/hr}}{20\ \text{L}} = 0.231\ \text{hr}^{-1}$ is the rate constant the later parts use.

> **DOES THE NUMBER MAKE SENSE?**
> Cl_T of 4.62 L/hr turns the 20 L volume over once in a little over 4 hours; a half-life is 0.693 of that turnover time, so about 3 hr fits.

b

Her key

Answer: 80 mg/L

WHAT THE STEM GIVES

weight	80 kg	cancels: dose and volume are both per kg
D_0	20 mg/kg	the numerator
τ	every 6 hours	*not needed for the first peak*
V_D	25% of body weight	0.25 L/kg, the denominator

Cl_T	4.62 L/hr	*not needed at time zero*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = \frac{D_0}{V_D} = \frac{20\ \text{mg/kg}}{0.25\ \text{L/kg}}$$

The first-dose peak is the dose over the volume of distribution; both are given per kilogram, so the weight cancels and the ratio can be taken directly.

ALGEBRA

$$C_0 = 80\ \text{mg/L}$$

mg/kg over L/kg leaves mg/L. The same value comes from the whole dose, (20 mg/kg)(80 kg) = 1600 mg, over the 20 L volume.

> **DOES THE NUMBER MAKE SENSE?**
> 20 mg in each 0.25 L is 4 × 20 = 80 mg/L; the same comes from the whole dose, 20 × 80 mg, in the 20 L volume.

C Her key

Answer: 20 mg/L

WHAT THE STEM GIVES

weight	80 kg	$V_D = 0.25 \times 80 = 20$ L
D_0	20 mg/kg	$C_0 = \frac{20}{0.25} = 80$ mg/L
τ	every 6 hours	the exponent $k\tau = 0.231 \times 6$
V_D	25% of body weight	20 L; $k = \frac{4.62}{20} = 0.231\ \text{hr}^{-1}$
Cl_T	4.62 L/hr	with V_D gives $k = 0.231\ \text{hr}^{-1}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{min} = C_0\, e^{-k\tau},\ k = \frac{Cl_T}{V_D} = \frac{4.62\ \text{L/hr}}{20\ \text{L}} = 0.231\ \text{hr}^{-1}$$

The trough of the first dose is the first-dose peak after one interval of first-order decline; the rate constant comes from clearance over volume, L/hr over L leaving hr^{-1}.

ALGEBRA

$$C_{min} = (80\ \text{mg/L})\, e^{-(0.231)(6)} = (80\ \text{mg/L})(0.25) = 20\ \text{mg/L}$$

The exponent $(0.231\ \text{hr}^{-1})(6\ \text{hr}) = 1.386$ is a pure number; 6 hours is two half-lives, so one quarter of the peak remains.

> **DOES THE NUMBER MAKE SENSE?**
> Six hours is two half-lives of 3 hours, so the first-dose peak halves twice: 80 mg/L to half, then to a quarter, 80 × 0.25 = 20 mg/L. The trough lies below the 80 mg/L peak.

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d

Her key

Answer: 106.68 mg/L

WHAT THE STEM GIVES

weight	80 kg	$V_D = 0.25 \times 80 = 20$ L
D_0	20 mg/kg	$C_0 = \frac{20}{0.25} = 80$ mg/L, the numerator
τ	every 6 hours	the exponent $k\tau = 0.231 \times 6$
V_D	25% of body weight	20 L; $k = \frac{4.62}{20} = 0.231$ hr^{-1}
Cl_T	4.62 L/hr	with V_D gives $k = 0.231$ hr^{-1}

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{max}^{\infty} = \frac{C_0}{1 - e^{-k\tau}}$$

The first-dose peak, 80 mg/L, divided by $(1 - e^{-k\tau})$ adds the drug left over from the earlier doses, so the steady-state peak must come out above 80 mg/L.

ALGEBRA

$$C_{max}^{\infty} = \frac{80 \text{ mg/L}}{1 - e^{-(0.231)(6)}} = \frac{80 \text{ mg/L}}{0.7499} = 106.68 \text{ mg/L}$$

$e^{-1.386} = 0.2501$ of each dose remains at the end of an interval, so 0.7499 is eliminated; dividing by it raises the peak by the accumulation factor 1.33.

> **DOES THE NUMBER MAKE SENSE?**
> A quarter of each dose remains after the 6 hour interval, so dividing by 0.7499 raises the 80 mg/L first-dose peak by about a third: near 107 mg/L. The steady-state peak must exceed the first-dose peak of 80 mg/L.

e

Her key

Answer: 26.68 mg/L

WHAT THE STEM GIVES

weight	80 kg	$V_D = 0.25 \times 80 = 20$ L
D_0	20 mg/kg	scaled to 1600 mg; $C_0 = 80$ mg/L, the numerator of the trough line
τ	every 6 hours	the exponent $k\tau = 0.231 \times 6$
V_D	25% of body weight	20 L; $k = \frac{4.62}{20} = 0.231$ hr^{-1}
Cl_T	4.62 L/hr	with V_D gives $k = 0.231$ hr^{-1}

THE WORKING, ONE LINE AT A TIME

UNIT

$D_0 = (20\ \text{mg/kg})(80\ \text{kg}) = 1600\ \text{mg}$; $V_D = 0.25 \times 80\ \text{kg} = 20\ \text{L}$; $C_0 = \dfrac{1600\ \text{mg}}{20\ \text{L}} = 80\ \text{mg/L}$

The dose and the volume both scale with the 80 kg weight, so C_0, the rise in concentration each dose produces, is the dose over the volume.

SETUP

$k = \dfrac{Cl_T}{V_D} = \dfrac{4.62\ \text{L/hr}}{20\ \text{L}} = 0.231\ \text{hr}^{-1}$; $k\tau = 0.231 \times 6 = 1.386$; $e^{-k\tau} = 0.2501$

The printed trough line wants k and τ; the stem gives clearance, so k comes from $Cl = kV_D$, and the 6-hour interval gives kτ, a pure number.

ALGEBRA

$$C_{min}^{\infty} = \frac{C_0\, e^{-k\tau}}{1 - e^{-k\tau}} = \frac{(80\ \text{mg/L})(0.2501)}{1 - 0.2501} = \frac{20.01}{0.7499} = 26.68\ \text{mg/L}$$

This is the printed trough line, read directly with C_0 from the stem: no peak is needed. Part d reaches the same number as the 106.68 mg/L peak times $e^{-k\tau}$.

DOES THE NUMBER MAKE SENSE?

Six hours is two half-lives, so a quarter of each dose is left when the next goes in; the trough is C_0 times 0.2501 over 0.7499, a third of the 80 mg/L C_0: near 26.7 mg/L, and below C_0.

f

Her key

Answer: 57.72 mg/L

WHAT THE STEM GIVES

weight	80 kg	$V_D = 0.25 \times 80 = 20$ L
D_0	20 mg/kg	$C_0 = \dfrac{20}{0.25} = 80$ mg/L, the numerator, with F = 1
τ	every 6 hours	$k\tau = 0.231 \times 6 = 1.386$ in the denominator
V_D	25% of body weight	20 L; $k = \dfrac{4.62}{20} = 0.231\ \text{hr}^{-1}$
Cl_T	4.62 L/hr	with V_D gives $k = 0.231\ \text{hr}^{-1}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{avg}^{\infty} = \frac{FD_0}{V_D\, k\, \tau} = \frac{C_0}{k\, \tau}, \text{ with } F = 1$$

D_0 over V_D is the first-dose peak, 80 mg/L, so the average at steady state is that peak divided by kτ; an IV dose is wholly available, so F = 1.

ALGEBRA

$$C_{avg}^{\infty} = \frac{80 \text{ mg/L}}{(0.231 \text{ hr}^{-1})(6 \text{ hr})} = \frac{80 \text{ mg/L}}{1.386} = 57.72 \text{ mg/L}$$

hr^{-1} times hr is a pure number, 1.386, so the unit stays mg/L; the same value comes from $\frac{1600 \text{ mg}}{(4.62 \text{ L/hr})(6 \text{ hr})}$. The average lies between the trough, 26.68, and the peak, 106.68 mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> $k\tau = 0.231 \times 6 = 1.386$, so the average is the 80 mg/L first-dose peak over 1.386, a little under 60 mg/L; it lies between the steady-state trough and peak and below their midpoint, because the decline is exponential.

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g

Her key

Answer: 1154 mg

WHAT THE STEM GIVES

weight	80 kg	$V_D = 0.25 \times 80 = 20$ L
D_0	20 mg/kg	the whole dose, $20 \times 80 = 1600$ mg, in $\frac{D_0}{k\tau}$
τ	every 6 hours	$k\tau = 0.231 \times 6 = 1.386$
V_D	25% of body weight	20 L, multiplies $C_{avg}^{\infty} = 57.72$ mg/L from part f
Cl_T	4.62 L/hr	with V_D gives $k = 0.231$ hr^{-1}

THE WORKING, ONE LINE AT A TIME

SETUP

$$D_{avg}^{\infty} = C_{avg}^{\infty} \times V_D$$

Amount and concentration are linked by the volume of distribution at every time, including on average at steady state; the average, 57.72 mg/L, is the one from part f.

ALGEBRA

$$D_{avg}^{\infty} = (57.72 \text{ mg/L})(20 \text{ L}) = 1154 \text{ mg}$$

mg/L times L: the litres cancel and an amount is left. The same result comes from the amount form, $\frac{D_0}{k\tau} = \frac{1600 \text{ mg}}{1.386} = 1154$ mg.

> **DOES THE NUMBER MAKE SENSE?**
> 57.72 mg/L in 20 L is a little under 60 × 20 mg, so 1154 mg fits; the average amount lies below the steady-state maximum amount, which exceeds one dose of 20 mg/kg × 80 kg.

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h

Her key

Answer: 6.67 mg/L

WHAT THE STEM GIVES

weight	80 kg	$V_D = 0.25 \times 80 = 20$ L

D_0	20 mg/kg	C_0 = 80 mg/L, inside C_{max}^{∞} = 106.68 mg/L from part d
τ	every 6 hours	*not needed: the time runs from the last dose*
V_D	25% of body weight	20 L; $k = \frac{4.62}{20} = 0.231\ hr^{-1}$
Cl_T	4.62 L/hr	with V_D gives $k = 0.231\ hr^{-1}$
t	12 hours after the last dose	the exponent $kt = 0.231 \times 12$

THE WORKING, ONE LINE AT A TIME

SETUP

$C = C_{max}^{\infty}\, e^{-kt}$, with t = 12 hr

After the last dose at steady state the level falls from the steady-state peak, 106.68 mg/L, by first-order elimination; the 12 hours is counted from the last dose, and the interval no longer enters.

ALGEBRA

$C12\infty = (106.68\ mg/L)\, e^{-(0.231)(12)} = (106.68\ mg/L)(0.0625) = 6.67\ mg/L$

The exponent $(0.231\ hr^{-1})(12\ hr) = 2.772$ is a pure number; 12 hours is four half-lives, so one sixteenth of the peak remains, below the steady-state trough of 26.68 mg/L because 12 hours is longer than the 6-hour interval.

> **DOES THE NUMBER MAKE SENSE?**
> Twelve hours is four half-lives of 3 hours, so one sixteenth (0.0625) of the 106.68 mg/L peak remains: about 6.7 mg/L. It lies below the steady-state trough because 12 hours exceeds the 6-hour interval.

Her key prints the exponent as $e^{-0.231(6)}$ with the result 6.67 mg/L; 6.67 is the 12-hour value, $e^{-0.231(12)}$. With 6 hours the value would be 26.68 mg/L, the steady-state trough.